

Questions from STEM SMART Students about NEA

Project Ideas & Scope

- Does it need to use a database? **No** (but it's a good way of getting better at SQL for the exam questions)
- Does it need a GUI graphical user interface? Can you just use CLI for your program? **It should use a GUI, but your teacher can contact the exam board in special circumstances.**
- Does a website using an Apache web server and a booking system using a database sound like a good idea? **Yes - although make sure there is some complex data processing happening in the background (e.g. scheduling appointments)**
- Is a Rubik's cube solver a valid NEA project? **Depends how you plan to model it? Might be too complex if you are trying to do it in 3D?**
- Is an artillery calculator for projectiles too simple? **Maybe - the stakeholders Maths/Physics students to help them learn SUVAT equations, so you could add a quiz at the end?**
- Would a multiplayer 1v1 website game be good (across multiple devices)? **Yes.**
- Could I build a study tool using the ClassCharts API? How would I start? **Look into a simple API Python library and work from there - getting Class Charts data might be a problem as it will have school/students private data in there? Avoid doing simple quizzes though - can be too basic for A Level.**
- Are there any good AI/machine learning/deep learning-related NEA ideas you could suggest? **ChatGPT is really good at suggesting project ideas...**
- What does it mean by robotics? Can you use a pre-created robot with your own code? **Depends - in theory yes, but the programming would need to be suitably complex. Not just Lego robotics!**
- Is it advisable to create a website or an app? **Avoid apps (unless you have built one already) There is a very steep learning curve for IDEs like Android Studio. A web site that can be accessed on a phone is much easier if you want the app experience)**
- What ideas would you recommend if it's not a game? **See slides!**
- What's one of the best projects you've seen in a NEA? **A contour map plotter that turned a 2D plot into a 3D representation of the terrain OR the program that took a photo of the night sky and could work out which area of the sky it was pointed**

at and identify the stars in the picture.

Tools, Platforms & Languages

- Where do I check which languages are allowed? Is Swift allowed in OCR? Is C++ allowed? Would JavaScript be okay? [Exam boards specifications and ask your teacher](#)
 - Can we use game engines? Will Unreal Engine be allowed? [Maybe - check with your teacher \(personally I would avoid as they abstract away much of the complexity you want to demonstrate\)](#)
 - Is Godot a good game engine? [Not sure - ask your teacher!](#)
 - What libraries can you use for Java/Python for games? [Almost all libraries are allowed - although you won't get credit for the code in the libraries, obviously.](#)
 - What should we use besides Tkinter for Python? [Guizero is good \(I'm a fan of Flask and use HTML/CSS for the front end\)](#)
 - Is Visual Studio Code good for the NEA? Specifically Python? [Yes!](#)
 - What is GitHub? [A website that you can store versions of your code on. It creates "backups" and tracks your changes. Recently teachers are likely to ask their A Level students to "push" their projects onto GitHub regularly so that they can monitor progress.](#)
-

Assessment & Specification

- What do you suggest should be included in each section (Analysis, Design, Testing) to get top marks? [See your exam boards specification.](#)
- Is there a brief for WJEC coursework? [See WJEC specification.](#)
- Do you need to have AI incorporated into your project? [No!](#)
- Are your total marks based on your own teacher? [Yes, but then a sample is sent to an external moderator.](#)
- How well does the program need to be optimised? [Depends - you can talk about how you would like to optimise it in your evaluation section, if you don't get round to doing it in the project.](#)
- Can we use ChatGPT to help us? [Check with your teacher. ALWAYS reference your use of it - NEVER try to pass off AI generated stuff as your own.](#)

- If we take code from somewhere online like Stack Overflow, should we credit it? Do we need to reference tutorials we use to help us code? [Put a list of links in a Reference section at the end \(or some teachers like you to paste the references throughout your project – check first\)](#)
 - How does API incorporation affect your score? Would it be worse for using external code, or better because it shows skill? [Only use an API if your project needs it!](#)
 - Which part of the project is awarded the most marks? Is it true that 70% of the NEA marks are given to the actual write-up and 30% for the code itself? [Depends on your exam board – look at your spec!](#)
-

Programming Concepts & Development

- Is it better to use OOP or procedural programming? What key coding concepts (e.g. OOP) do you recommend using in the project to get the best marks? [Use the concepts that make the most sense in your project. Don't use something just for the sake of it.](#)
 - Can you do multiple things in one project, like performance management and new game mechanics? [Yes... but it's often better to do one thing REALLY well than spread yourself too thin trying to do lots of things.](#)
 - What languages or types of projects would help someone be more equipped for a computing degree or industry job? [If you're looking to the future then I'd still recommend Python \(or C#\) as that's an industry standard \(not VB!\) Maybe avoid games? Look at projects that use lots of maths or data might be well regarded post A Level. But don't worry too much about it now – just pick an idea you're interested in and can do!](#)
-

Student Experience & Strategy

- What if you're not very good at coding? What would be best for someone with limited coding skills (only what's taught for the exam)? [You can score something for a project if you just write a good Analysis and Design. Or you could aim for an easier project with less complex skills.](#)
- Where is the best place to learn code? [Lots of good tutorials online – a doc was sent out on Moodle with links to lots of Python resources](#)
- Which is more important: making something fun like a game, or being strategic and “serious” for the marks? [It's more important to be within your skillset and something that you are interested in and will finish on time .](#)

- Do we have to finish our whole game at the end of the NEA? Ideally, but you can submit a write up that just includes Analysis , Design and some development to get some marks.
- I am stronger at coding than writing the report — is it still possible to get an A* with good code and a decent report? The write-up is the thing that is being marked so it is really important you spend time on the Analysis and Design sections before you begin to code in earnest. Fantastic code with no write-up (or poor objectives) will score next to nothing!
- If you overexplain some bits of simple code, will the exam boards just assume that you have no idea what to do? Probably! But if you've written something there is some chance you will get some marks.